Finitely generated unconfined Steiner triple systems are free

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Editorial note. The finite HF-ordering argument supplies nonnegative predimension by assigning each block to its greatest vertex, with at most one block per vertex. The cited companion paper is not granted as a hidden freeness theorem.

Editorial note. The strict-order compactness display294 lacks a!=b in its totality clause. Only the finite peeling direction of this support proposition is needed here; the infinite compactness display is retained as source context, not used to infer the target.

Editorial scope. One original proposition with its full author local proof chain; no new claim assigned to any support lemma.

Allowed standard prerequisites (editorial).

- Elementary universal algebra and well-founded recursion. Universal mapping property, generated subalgebras, uniqueness of a homomorphism on generators, and recursion/induction on well-orders.
- Finite combinatorial accounting on terms and triples. Term length and rank, counting blocks by their greatest vertex, and monotone nonnegative integer sequences.

A **Steiner triple system** (STS) of order λ is a pair $(M, \mathcal{B})$ where M is a set with λ elements, and $\mathcal{B}$ is a collection of 3-element subsets of M (the **blocks**) such that any two $x, y \in M$ are contained in exactly one block. If we require that any two $x, y \in M$ are contained in *at most* one block, M is said to be a **partial Steiner triple system**.

From a model theoretic standpoint, Steiner triple systems can be viewed as relational structures with a single a ternary relation R interpreted as the set of blocks (that is, three elements x, y and z are in a block if and only if $R(x, y, z)$). Then R must satisfy

- for all permutations $\sigma \in S_3$, $\forall x_1 x_2 x_3 \ \big(R(x_1, x_2, x_3) \to R(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)})\big)$
- $\forall xyz \, (R(x, y, z) \to y \neq z)$
- $\forall xywz \, (R(x, y, w) \wedge R(x, y, z) \to w = z)$.

With this choice of language, a substructure of a Steiner triple system is a partial Steiner triple system.

An STS $(M, \mathcal{B})$ determines a *Steiner quasigroup*, that is, a pair $(M, \cdot)$ where $\cdot$ is a binary operation such that

- $\forall x\,(x \cdot x = x)$
- if $x \neq y$, then $x \cdot y = z$ where $\{x, y, z\} \in \mathcal{B}$.

The operation $\cdot$ defined in this way satisfies

1. $\forall x\,(x \cdot y = y \cdot x)$
2. $\forall x\,(x \cdot x = x)$
3. $\forall x\,(x \cdot (x \cdot y) = y)$.

Conversely, any structure $(M, \cdot)$ where $\cdot$ satisfies properties 1, 2 and 3 determines an STS $(M, \mathcal{B})$.

In universal algebraic parlance, Steiner quasigroups are algebras. Thus, if M is a Steiner quasigroup, there are standard notions of a *subalgebra* of M, and of the subalgebra *generated* by a subset $A \subseteq M$. A subalgebra of a Steiner quasigroup is a substructure in the model theoretic sense, and in particular it is a Steiner quasigroup. If $A \subseteq M$, then $\langle A \rangle_M$ denotes the substructure of M generated by A. The subscript M is usually omitted when clear from the context.

A Steiner quasigroup induces a partial STS structure on each of its subsets, so we often refer to a partial Steiner triple system when we need to consider a subset of a Steiner quasigroup that is not a substructure.

Given a class $\mathcal{V}$ of algebras in a given signature, we say that a member U of $\mathcal{V}$ generated by a subset $X \subseteq U$ has the **universal mapping property** (UMP) over X with respect to $\mathcal{V}$ if for every $V \in \mathcal{V}$, every map

$$\varphi : X \to V$$

extends to a homomorphism $\hat{\varphi} : U \to V$. A standard argument proves that the homomorphism $\hat{\varphi}$ is unique. Such an algebra U is said to be **freely generated by** X, and X is a **set of free generators** of U.

It is well known that when $\mathcal{V}$ is a variety, given any cardinal $\lambda \geq 1$ there are $U \in \mathcal{V}$ and $X \subseteq U$ such that U is freely generated by X and $|X| = \lambda$.

Since the class $\mathcal{K}$ of Steiner quasigroups is a variety in the universal algebraic sense, for each cardinal $\lambda \geq 1$, there is a Steiner quasigroup that is freely generated by a set of cardinality λ. Then a **free Steiner quasigroup** is a Steiner quasigroup that is freely generated by some subset X, and so it has UMP over X with respect to the class $\mathcal{K}$. We assume $\lambda \geq 3$ to exclude the trivial cases of Steiner quasigroups with one and three elements.

Definition 0.1 Let $(M, \cdot)$ be a Steiner quasigroup. A subset $A \subseteq M$ is **independent** if the generated substructure $\langle A \rangle_M$ is freely generated from A, that is, if $\langle A \rangle_M$ has UMP over A for the class of Steiner quasigroups.

We say that $A \subseteq M$ is a **free base** from M if it is independent and it generates M.

When we say $\bar{a} = \langle a_i : i \in I \rangle$ is independent, we understand $a_i \neq a_j$ if $i, j \in I$ and $i \neq j$.

If M is a Steiner quasigroup generated by the subset A and $\bar{a}$ is an enumeration of A, then for every $b \in M$ there is a term $t(\bar{x})$ such that $b = t(\bar{a})$.

Let M be a Steiner quasigroup generated by $A \subseteq M$. Then $M = \bigcup_{n<\omega} S_n$ for a chain of sets S_n such that $S_0 = A$ and $S_{n+1} = \{a \cdot b \mid a, b \in S_n\}$. Note that $S_n \subseteq S_{n+1}$ since $a \cdot a = a$. We will write $S_{<n}$ for $\bigcup_{i<n} S_i$. The sets S_n are called the **levels** of M over A. An element $a \in M$ has **level** n **over** A if $a \in S_n \setminus S_{<n}$.

Definition 0.2 A partial STS A is **unconfined**, or **contains no confined configuration**, if for all finite non-empty $A' \subseteq A$ there is $a \in A'$ that belongs to at most one block in A', that is, a is the product of at most two elements of A'.

Definition 0.3 Let A be a finite partial STS and let $\mathrm{b}(A)$ be the set of blocks in A. Then the **rank**, or **predimension**, of A is

$$\delta(A) = |A| - |\,\mathrm{b}(A)|\,.$$

It is easy to see that the predimension of a finite Steiner quasigroup is negative except in the case of the Steiner quasigroup of order 7 (the Fano plane), which has predimension 0. On the other hand, the predimension of a finite unconfined partial Steiner triple system is always non-negative (see [2] for details).

Let Aa be a partial Steiner triple system, with $a \notin A$. Then

$$\delta(Aa) = \delta(A) + 1 - |\,\mathrm{b}_a(A)|\,,$$

where $\mathrm{b}_a(A)$ is the set of blocks containing a and intersecting A. Hence, if there is at most one such block, $\delta(Aa) \geq \delta(A)$. It follows that if $A \subseteq B$ are finite partial Steiner triple systems and there is an HF-ordering of B with initial segment A, then $\delta(B) \geq \delta(A) \geq 0$.

Proposition 0.4 Finitely generated unconfined Steiner triple systems are free Steiner quasigroups.

Proposition 0.5 Let M be a Steiner quasigroup and let $A \subseteq M$. The following are equivalent:

1. A is a free base of M.
2. M is generated by A with levels $(S_n \mid n < \omega)$ such that $A = S_0$ and
   (a) if $a, b \in S_n \setminus S_{<n}$ and $a \neq b$, then $a \cdot b \notin S_n$.
   (b) if $a \in S_{n+1} \setminus S_n$, there is a unique pair $\{b, c\} \subseteq S_n$ such that $b \neq c$ and $a = b \cdot c$.

Moreover, property (b) implies that if $a \in S_{n+1} \setminus S_n$, and $b, a \cdot b \in S_n$, then $\{b, a \cdot b\}$ is the unique pair given by (b) for a.

Proof. $2 \Rightarrow 1$. Let N be a Steiner quasigroup and let $f : A \to N$ be a mapping. We extend f to a homomorphism $\hat{f} : M \to N$. We obtain $\hat{f}$ as the union $\hat{f} = \bigcup_{n<\omega} f_n$ of a chain of partial homomorphisms $f_n : S_n \to N$. This means

that $f_n(a \cdot b) = f_n(a) \cdot f_n(b)$ whenever a, b and $a \cdot b$ belong to $\mathrm{dom}(f_n) = S_n$. We start with $f_0 = f$. Recall that $A = S_0$ and that $a \cdot b \notin S_0$ if $a, b \in S_0$ and, therefore, f_0 is a partial homomorphism with domain S_0. Now we extend $f_n : S_n \to N$ to f_{n+1}. For $a \in S_{n+1} \setminus S_n$, there is a unique pair $\{b, c\} \subseteq S_n$ of distinct elements b, c with $a = b \cdot c$. We let $f_{n+1}(a) = f_n(b) \cdot f_n(c)$. By (a) and (b), the only defined products involving a in S_{n+1} are $a = b \cdot c = c \cdot b$, $a \cdot a = a$, $a \cdot b = b \cdot a = c$ and $a \cdot c = c \cdot a = b$ and f_{n+1} respects these products.

$1 \Rightarrow 2$. It is easy to construct a Steiner quasigroup $M' \supseteq A$ with the properties described in 2. It is enough to start with $S'_0 = A$ and inductively define S'_{n+1} by adding to S'_n a different element $a \cdot b$ for each pair $\{a, b\}$ of elements of S'_n whose product is not defined in S'_n. The only products involving the new element $c = a \cdot b$ in S'_{n+1} are $c = a \cdot b = b \cdot a$, $c \cdot c = c$, $c \cdot a = a \cdot c = b$ and $c \cdot b = b \cdot c = a$. We have proven that A is a free base of $M' = \bigcup_{n<\omega} S'_n$. The uniqueness of free Steiner quasigroups over a given set of generators guarantees that M and M' are isomorphic over A. The isomorphism can be used to obtain the levels S_n in M with the required properties. □

Definition 0.6 A **free ordering** (or an **F-ordering**) of a Steiner quasigroup M over $A \subseteq M$ is a linear ordering $<$ of M such that for every $a \in M$ there is a unique pair $\{a_1, a_2\} \subseteq M$ with $a = a_1 \cdot a_2$, and for each i, either $a_i \in A$ or $a_i < a$.

Note that, when $a \in A$, the unique pair is given by $a_1 = a_2 = a$.

Remark 0.7 Let M be a Steiner quasigroup with standard free construction $(S_n \mid n < \omega)$ over A. Let $<_n$ be a linear ordering of $S_n \setminus S_{<n}$, and for $a, b \in M$ define

$$a < b \Leftrightarrow \text{ for some } n,\ b \in S_n \setminus S_{<n} \text{ and either } a \in S_{<n} \text{ or } a \in S_n \setminus S_{<n} \text{ and } a <_n b.$$

Then $<$ is an F-ordering of M over $A = S_0$, and S_0 is an initial segment. If every $<_n$ is a well-order, then $<$ is a well-order.

Remark 0.8 Let M be a Steiner quasigroup. If there is an F-ordering of M over A, there is another one where A is an initial segment. If the original order is a well-order, the new one is also a well-order and, moreover, $M = \langle A \rangle$.

Definition 0.9 A **hyperfree ordering** (or **HF-ordering**) of a partial Steiner triple system A is a linear ordering $<$ of A such that for every $a \in A$ there is at most one pair of smaller elements $\{a_1, a_2\} \subseteq A$ such that $a = a_1 \cdot a_2$, that is, there is at most one block $\{a, a_1, a_2\}$ with $a_1, a_2 < a$.

The next proposition characterises free bases in terms of the existence of specific F-orderings and HF-orderings. Recall that a well-founded linear order is a well-order.

Proposition 0.10 Assume M is a Steiner quasigroup and $A \subseteq M$. The following are equivalent:

1. A is a free base of M.
2. There is an F-ordering of M over A which is a well-order.

3. There is an HF-ordering of M which is a well-order and A is the set of points that are not a product of smaller elements.

Proof. $1 \Rightarrow 2$. By Proposition 0.5, there is standard free construction $(S_n \mid n < \omega)$ of M over A. Choose a well-order $<_n$ of every set $S_n \setminus S_{<n}$. By Remark 0.7, there is an F-ordering $<$ of M over A which is a well-order extending all orders $<_n$.

$2 \Rightarrow 3$. By Remark 0.8 there is an F-ordering $<$ of M over A which is a well-order and A is an initial segment. We know that no element of A is a product of smaller elements. If $a \in M \setminus A$, then $a = a_1 \cdot a_2$ where each a_i is either smaller than a or an element of A. Since A is an initial segment, $a_1, a_2 < a$.

$3 \Rightarrow 1$. Let $<$ be a well-order and an HF-ordering of M and assume A is the set of elements which are not a product of smaller elements. As in Remark 0.8, we can modify the order and make A an initial segment. Hence, every element of $M \setminus A$ is a product of smaller elements. By induction on the well-order it is easy to see that $M = \langle A \rangle$. Note that no element of A is a product of two different elements of A. Let N be a Steiner quasigroup and $f : A \to N$ a mapping. By recursion on the well-order, it is easy to extend f to some homomorphism $f' : M \to N$. □

Proposition 0.11 The following are equivalent for any partial Steiner triple system A:

1. A is unconfined.

2. There is an HF-ordering of A.

Proof. Consider first the case A finite. By induction on $|A|$ it is easy to prove that, if A is unconfined, then it has an HF-ordering. On the other hand, if there is an HF-ordering of A, for any non-empty $A' \subseteq A$ the greatest element $a \in A'$ belongs to at most one block of A, hence to at most a block of A'. Consider now the general case. If A has an HF-ordering, every finite subset of A has an HF-ordering and so it is unconfined, which implies that A is unconfined. For the other direction, we use propositional compactness. Assume that every finite subset of A has an HF-ordering. For every pair $(a, b) \in A \times A$, let p_{ab} be a corresponding propositional variable. The set of sentences

$$\{p_{ab} \wedge p_{bc} \to p_{ac} \mid a, b, c \in A\} \cup \{\neg p_{aa} \mid a \in A\} \cup \{p_{ab} \vee p_{ba} \mid a, b \in A\}$$

defines a linear ordering on A. If we add

$$\{\neg(p_{ba} \wedge p_{ca} \wedge p_{b'a} \wedge p_{c'a}) \mid \{a, b, c\} \neq \{a, b', c'\} \text{ blocks of } A\}$$

we are defining an HF-ordering on A. Every finite subset is satisfiable and compactness gives a valuation that satisfies the whole set. □

Proof. Let A be a finite set of generators of the unconfined Steiner quasigroup M. Let $(S_n \mid n < \omega)$ be the levels of M over A. Every S_n is finite. Each element of $S_{n+1} \setminus S_n$ is in at least one block intersecting S_n, and hence there is an enumeration $(a_i \mid i < k)$ of $S_{n+1} \setminus S_n$ such that $\delta(S_n a_1, \ldots, a_{i+1}) \leq \delta(S_n a_1, \ldots, a_i) \leq \delta(S_n)$. Concatenating these enumerations after an arbitrary enumeration of $A = S_0$ gives an enumeration $(a_n \mid n < \omega)$ of M. Since M is

unconfined, $\delta(a_0, \ldots, a_n) \geq 0$ for every finite n. Therefore there is some n such that, for every $m \geq n$, $\delta(a_0, \ldots, a_m) = \delta(a_0, \ldots, a_n)$. Since M is unconfined, $A_m = \{a_i \mid i \leq m\}$ has an HF-ordering, and we may assume that the elements of A_m which are not a product of smaller elements form an initial segment $A'_m \subseteq A_m$. We start with this HF-ordering of A_m and add the enumeration $(a_i \mid i > m)$, obtaining an F-ordering of M over A'_m which is a well-order. Then A'_m is a free base of M by Proposition 0.10.

□

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